NAME:_____ Time Start:_____ Time End:_____

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Question 9

Show your workings clearly and round off your answers to ***two decimal places*** ***where necessary.*** *Use ^ symbol to represent the power notation.*

At a self-service ordering kiosk, an average of 15 customers arrive every hour. The time each customer takes to place their order at the kiosk is a mean of 3 minutes with a standard deviation of 1 minute.

Question 9a (3 marks)

Using Kendall's notation, state a suitable queuing model for the above queuing system. List 2 assumptions of the queuing model.

M/G/1.

2 assumptions:
- Queue discipline: FIFO (FCFS)
- Service mechanism: Single server, single phase system

Question 9b (4 marks)

Calculate:
i. Mean arrival rate of customers in per minute (2 marks)
ii. Utilization of the kiosk (2 marks)

i. Mean arrival rate = 15/60 = 0.250 per min

ii. Service rate = 1/3 = 0.333 per min
Utilization = 0.250/0.333 = 0.75

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Question 9c (4 marks)

Calculate:
i. The average number of customers waiting at the kiosk, Lq. (2 marks)
ii. The average time customers spend in the system, Ws in minutes. (2 marks)

i. Lq = ((0.250^2*1^2)+0.750^2)/(2*(1-0.750)) = 1.25

ii. Wq = Lq/Arrival rate
= 1.25/0.250
= 5 mins

Ws = Wq + 1/Service Rate
= 5 + (1/0.333)
= 8 mins

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Question 9d (3 marks)

Suppose the number of customer arrivals has increased to 90 per hour. Will the queuing system still be in steady state? Determine the minimum number of kiosks required if the mean service time remains at 3 minutes.

The queuing system will not be in steady state.

NEW Arrival rate = 90/60 = 1.5 per min
Service rate = 1/3 = 0.33 per min

Steady state equilibrium holds when arrival rate < k*Service rate(for k kiosks)

Therefore, minimum k > arrival rate/service rate > 1.5/0.33 > 4.5

Minimum number of kiosks required, k = 5

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Question 9e (2 marks)

State the following characteristics (assumptions) in order that a M/M/k queuing model can be applied to study a queuing system with multiple kiosks.

i. Service time distribution _____

ii. Single queue or multiple queue lines _____

I. Negative exponential distribution

II. Single Queue

Question 9f (2 marks)

Given that the average number of customers in queue in the M/M/k model, Lq is 6.86, calculate the average waiting time of customers in the queue, Wq in minutes (assume arrival rate of 90 per hour).

Arrival rate = 90/60 = 1.5 per min

Wq = Lq/Arrival rate
= 6.86/1.5
= 4.57 mins